ABHINANDAN MANDAL

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Research Focus

Efficient and interpretable visual perception and reasoning, what models actually represent, reason, how much capacity they need, and where they fail. Applied work in reinforcement learning, out-of-distribution and parallel algorithms for efficient computation.

Education

Indian Institute of Information Technology and Management (IIITM), Gwalior Aug. 2025 – Present
M.S. in Artificial Intelligence & Data Science **CGPA: 9.36/10**
Coursework: Computational Linear Algebra, Artificial Intelligence, Deep Learning, NLP, Advanced Graph Theory, Numerical Methods for Data Science

Rajdhani College, University of Delhi Nov. 2021 – July 2024
B.Sc. (Honours) Physics **CGPA: 7.52/10**
Coursework: Quantum Mechanics, Nuclear & Particle Physics, Classical Dynamics, Statistical Mechanics, Advanced Mathematical Physics

Publications

- **A. Mandal**, S. Kumari, Y. S. Patel, A. Pratap. "An Adaptive Attention-Based Improved Double Deep Q-Network Approach for Robust and Energy-Aware Autonomous Mobile Robot Navigation in GNSS-Denied Environments." **ICDCN 2027**. GitHub
- G. Manivannan, **A. Mandal**. "Bridging Pretrained Vision and Symbolic Reasoning Through Intermediate-Layer Grounding." **ACCV 2026** Workshop (MM4SciReal). GitHub
- **A. Mandal**, Y. S. Patel, A. Choubey, A. Singh. "MoSE-XAI: A Lightweight MobileNetV2 Squeeze-and-Excitation Model for Crop Disease Detection with XAI Visualization." **COMSNETS 2026**, Graduate Forum. GitHub
- **A. Mandal**, Y. S. Patel. "An Energy-Feasible Parallel Algorithm for Large Dynamical Networks." **ICIIS 2026**. GitHub

Research Experience

Summer Research Intern, IIT (BHU) Varanasi May 2026 – July 2026
Advisor: Dr. Ajay Pratap

- Built **A-IDDQN**, a deep-RL navigation policy for GNSS-denied mobile robots, an adaptive attention module reweights LiDAR, camera, and IMU inputs per navigation state, and an energy-utility term shapes the reward across navigation and environments. GitHub
- Outperformed baseline DQN, DDQN, Dueling-DQN, and IDDQN by **7.88%** in navigation efficiency with **11.11%** shorter paths; gains attributed to the attention module and energy-utility modeling with improved double deep-Q network.

Graduate Research Intern, IIITM Gwalior Sept. 2025 – Dec. 2025
Advisor: Dr. Yashwant Singh Patel

- Designed **MoSE-XAI**, a MobileNetV2 architecture with Squeeze-and-Excitation blocks for crop-disease detection, cutting parameters **27.65%** versus the MobileNetV2 baseline at **98.96%** accuracy, with Grad-CAM visualizations for interpretability. GitHub

Research Projects

Rank-Constrained Semantic Segmentation via Sparse-Semantic Factorization *extends STELLAR (ICML 2026)* GitHub

- Tested whether dense features are over-parameterized by sweeping intrinsic rank $\in \{8, 15, 16, 32, 64\}$ on GlaS histopathology segmentation. Rank-32 (80.43% compression) matched the dense baseline (Dice 0.8457 ± 0.0031 vs. 0.8437 ± 0.0034); rank-15 (90.29% compression) stayed within 0.004 Dice (0.8399 ± 0.0048), suggesting redundant capacity in the dense model.
- Accepted at the AI+Science Conference 2026 (UChicago & Caltech). Ongoing: testing the "task-aligned intrinsic-dimensionality hypothesis" across biological datasets.

WildCountBench: Interpreting Visual Counting Bottlenecks in Multimodal Models *extends NegBench, CountQA (CVPR 2025)* GitHub

- Benchmarked CLIP-based VLMs on NegBench: **80%** false-positive rate on negated prompts vs. **40%** on affirmative ones, consistent with contrastive objectives not modeling negation.
- In progress: collecting CountQA-style in-the-wild counting data to localize failures with mechanistic interpretability.

Pose-Invariant Multi-View Object Representation in Latent Space *extends PiRO (CVPR 2024)* GitHub

- Reproduced the PiRO dual-embedding baseline; ablations separating category- from object-level embeddings showed object-level embeddings drive most of the gains: $+4.5\%$ and $+10.9\%$ classification, $+1.5\%$ and $+6.4\%$ retrieval on ModelNet-40 and ObjectPI.
- In progress: curating a multi-view everyday-object dataset to test cross-category generalization.

Technical Skills

Core: Python, PyTorch, C++, TensorFlow **Experimentation:** Weights & Biases, scikit-learn, NumPy, Pandas, Matplotlib **Tooling:** Git, Docker, FastAPI

Honors, Service & Leadership

- **Best Poster Presentation**, Gwalior AI Summit 2026. **Selected participant:** GCI World 2026, Stanford Probability for AI.
- **Peer reviewer:** NeurIPS GlobalSouthAI 2026, NeurIPS RTCA 2026, ACCV MM4SciReal 2026.
- **Research Lead**, Advanced Vision Labs; former Community Lead, ML Paper Club; JEPA research cohort contributor, The Bu1LD.